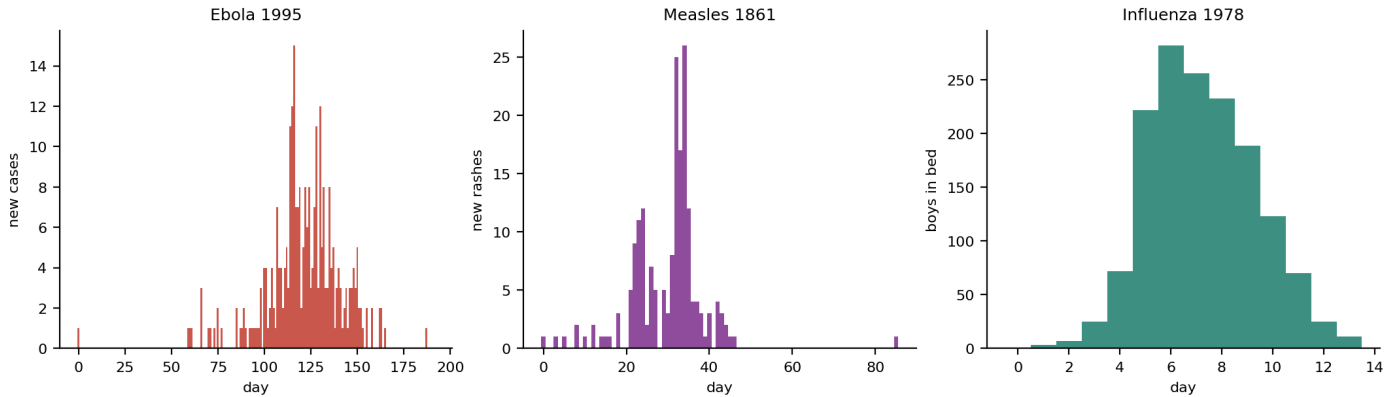


Four Outbreaks: What the Checks Found

Everything below was computed from the raw records before any model was fitted

Outbreak	Where and when	Who	Caught it	Died of it	Records
Ebola	Kikwit, DR Congo, 1995	~1,000 people	292	81%	192 days
Measles	Hagelloch, Germany, 1861	188 children	188 (all)	6.4%	86 days
Influenza	English boarding school, 1978	763 boys	512	0%	14 days
Norovirus	Derbyshire school, England, 2001	492 children	75	~0%	492 children

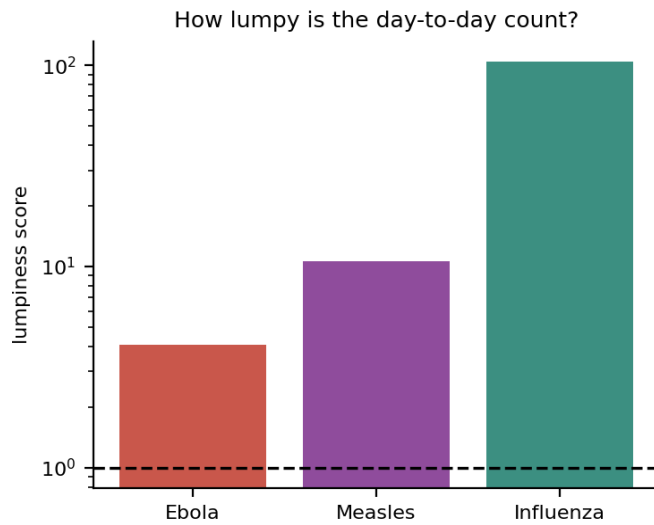
This shows that these four span the full range of what an infectious disease can be: one that kills most people it infects but spreads slowly, one that infects absolutely everyone but rarely kills, and two in between. That contrast is the point of studying them together.



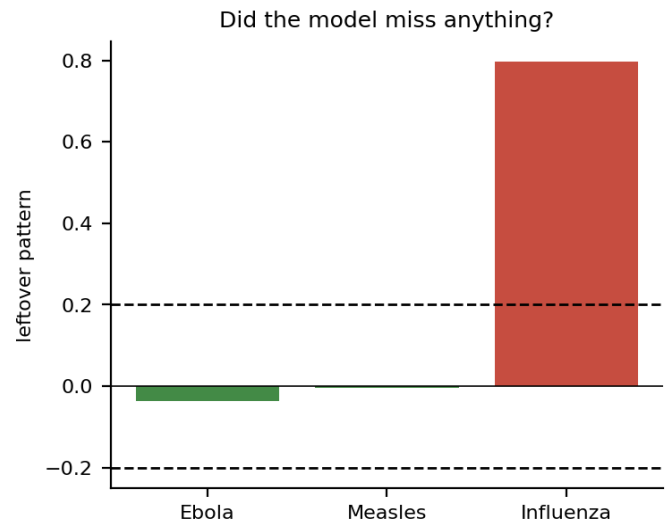
This shows that each outbreak rose once and fell once. That single-hump shape means each burned through its group and stopped, rather than smouldering on — which is what makes them comparable to one another at all. Note the horizontal scales differ: measles took three months, flu took two weeks.

The Four Checks

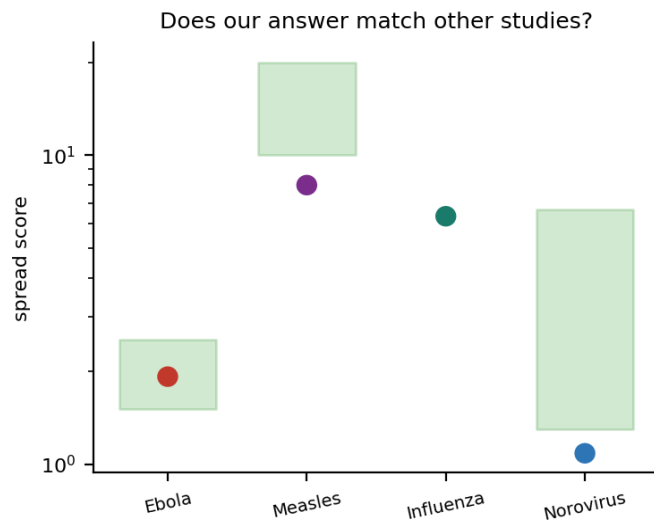
Each one decides something concrete about how the outbreak is modelled



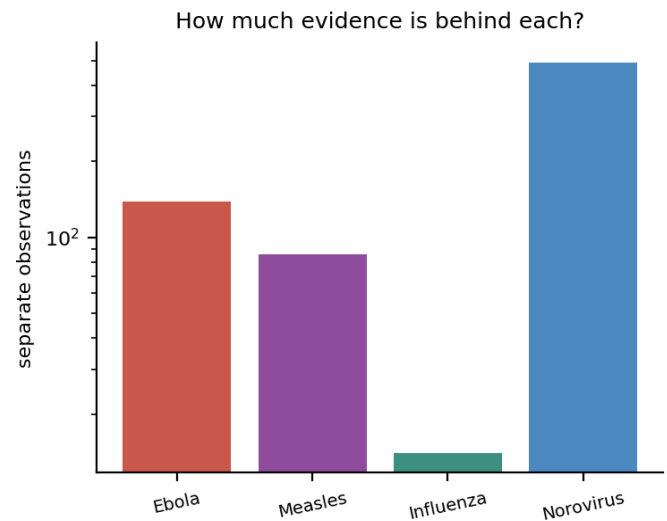
This shows that day-to-day counts jump around far more than a simple tally would predict — all three sit well above the dashed line. So we use a more forgiving way of counting that expects quiet days followed by clusters.



This shows that for Ebola and measles the model left nothing behind (green). For influenza it did (red). That single red bar is why influenza is treated differently from the other three.



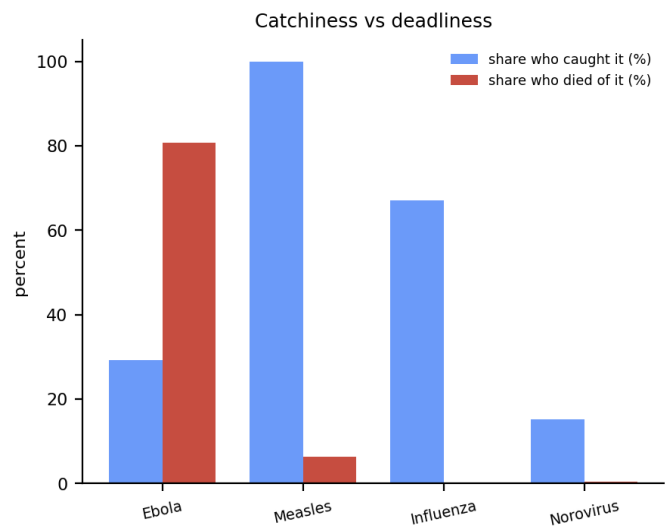
This shows that the green bands are what other researchers have published for each disease. Ebola and norovirus land inside theirs. Measles lands a little below, which we expect because our model assumes a village mixes evenly and villages with schools do not.



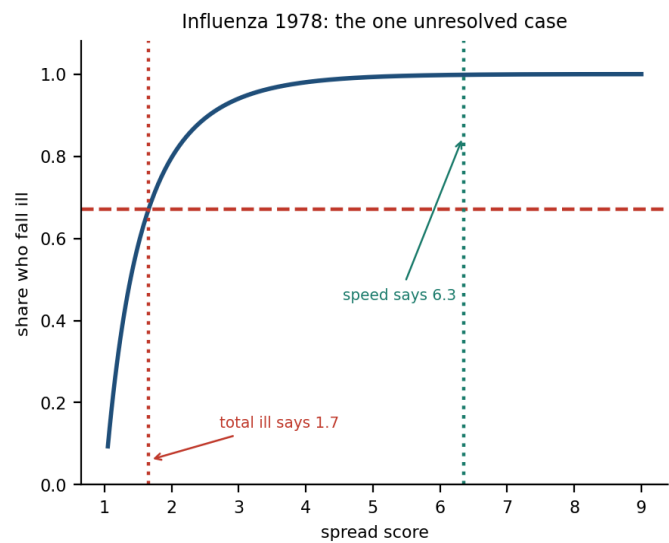
This shows that norovirus rests on nearly 500 individual children; influenza rests on 14 daily tallies. More evidence means a narrower, more trustworthy answer — this is the second reason influenza is the weakest of the four.

What the Numbers Mean in Plain Terms

Two comparisons worth showing a non-specialist directly



This shows that catchiness and deadliness are separate things. Measles infected every child in the village but killed one in sixteen of them. Ebola infected a few hundred people but killed four in five. Neither number alone tells you how dangerous an outbreak is.



This shows that for influenza the two clues point to different answers and there is no way to satisfy both at once. Published attempts on this same dataset run into the identical wall, so this is a property of the records, not a mistake in our work.

Decisions that follow directly from the checks

Outbreak	What was counted	How we count it	Extra time allowance?	Verdict
Ebola	new cases each day	forgiving count	not needed	PASS
Measles	new rashes each day	forgiving count	not needed	PASS
Norovirus	ill or not, per child	per-child, per-classroom	not applicable	PASS
Influenza	boys in bed each day	forgiving count	would swamp the data	CAVEAT

This shows that three of the four needed no special handling at all. Influenza needed an extra allowance for things changing over time, but with only 14 days of records that allowance would absorb everything and leave nothing to measure — so it is left out and the limitation is stated instead.

How to Read the Results When They Arrive

Four things to check, in order

Step	What to look at	What good looks like
1	Ebola, measles and norovirus contagiousness estimates	each range overlaps the published range for that disease
2	The range, not the single number	a wide range is an honest answer from thin records, not a failure
3	Influenza quoted only with its caveat	never placed beside the other three as an equal
4	The run repeated without influenza	the other three barely move, confirming it is not distorting them

This shows that the headline claim of this work rests on ebola, measles and norovirus. Influenza is included to show the checks can catch a bad case, not to carry an argument.

One-line summary of each outbreak

Outbreak	In one sentence
Ebola	Spread slowly and killed most people it infected; our model fits it cleanly.
Measles	Infected every child in the village; model fits the timing, smooths the classroom bursts.
Norovirus	Mild, clustered in classrooms, and the best-recorded of the four.
Influenza	Fastest and mildest, but the records give two contradictory answers; kept as a test case.

This shows that if the professor reads only one thing on this page, it is this table.

Reference Numbers

The same checks in their technical form, for anyone who wants them

Outbreak	Var/Mean	Skew	Zero-day share	Residual lag-1	Ljung-Box p	Const-fit R0	GRW warranted
Ebola	4.06	1.95	0.41	-0.0374	0.8847	1.92	no
Measles	10.64	3.31	0.59	-0.0047	0.7310	8.01	no
Influenza	103.64	0.45	0.00	+0.7961	0.0033	6.35	YES
Norovirus	n/a	n/a	n/a	n/a	n/a	1.08	n/a

This shows that this is the same information as the coloured charts, written the way a statistician would want it. Nothing new is here — it is included so the numbers behind every claim can be checked.

Vocabulary, once

Term used here	What it means
spread score (R0)	how many people one infected person passes it to, on average, in a group with no immunity
lumpiness (Var/Mean)	how much more the daily counts jump around than a simple tally would predict
leftover pattern	whether the model's errors still follow a trend after it has done its best
share who fall ill	the fraction of the whole group that eventually catches it
forgiving count	a way of counting that expects quiet days followed by clusters, rather than a steady trickle

This shows that these five terms cover every chart in this report.